

Units

$N = \frac{kg \cdot m}{s^2}$
 $R = \frac{J}{kg \cdot K}$
 $\rho = \frac{kg}{m^3}$
 $\mu = \frac{kg}{m \cdot s}$
 $\gamma = \frac{C_p}{C_v}$
 $\delta = \frac{g}{s}$ (l/s or m/s)

Laws of Classical Thermodynamics

1st: $\Sigma Q = \Sigma W$
 $Q = W + \Delta E$
 $Q = W + \Delta e$
 $e = u + \frac{V^2}{2gc} + \frac{g}{gc} z$
 $dh = du + dP/\rho = du + P dV/\rho + V dp$
 $C_p = \left(\frac{\partial h}{\partial T}\right)_P \Rightarrow dh = C_p dT$
 $C_v = \left(\frac{\partial u}{\partial T}\right)_V \Rightarrow du = C_v dT$
 $\gamma = \frac{C_p}{C_v}$, $C_p - C_v = \frac{R}{M}$
 $\approx C_p - C_v = R$

Equation of state: $P = \rho R T$ Perfect Gas

Mass flow rate $\dot{m} = \int \rho V dA$, $\dot{m} = \rho A V = P A \frac{V}{R T}$

Continuity

1-D, $\dot{m} = \rho A V = C$ $\frac{dP}{P} + \frac{dA}{A} + \frac{dV}{V} = 0$

Reynolds Transport

$\frac{dN}{dt} = \frac{d}{dt} \int_{CV} \rho \phi dV + \int_{CS} \rho \phi (\vec{V} \cdot \vec{n}) dA$
 Mass $\phi = 1$ $\frac{dN}{dt} = \frac{dM}{dt} = 0$
 Mass $\phi = V$ $\frac{dN}{dt} = \frac{dV}{dt} = \Sigma F$
 URG $\phi = e$ $\frac{dN}{dt} = \frac{de}{dt} = \frac{dQ}{dt} - \frac{dW}{dt}$

1-D, $h_1 + \frac{V_1^2}{2gc} + \frac{g}{gc} z_1 = h_2 + \frac{V_2^2}{2gc} + \frac{g}{gc} z_2 + w_s$

$\Rightarrow \delta q = \delta w_s + dh + \frac{dV^2}{2gc} + \frac{g}{gc} dz$

Entropy division $ds = ds_c + ds_i = \frac{\delta q}{T} + ds_i$

Pressure-energy Eq $\frac{dP}{\rho} + \frac{dV^2}{2gc} + \frac{g}{gc} dz + \delta w_s + ds = 0$

Stagnation Concept $h_t = h + \frac{V^2}{2gc} + \frac{g}{gc} z$

Stagnation Pressure Energy: $\frac{dP_t}{\rho} + ds_c(T_t - T) + T_t ds_i + \delta w_s = 0$

if $\delta w_s = 0$ + No losses $P_t = \text{const}$

Constant density fluids $\frac{\Delta P}{\rho} + \frac{\Delta V^2}{2gc} + \frac{g}{gc} \Delta z = h_L + w_s$

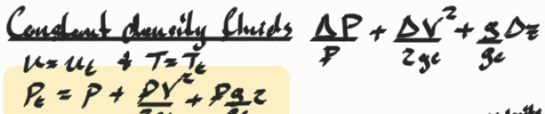
Second Law - Momentum $\Sigma F = \frac{d}{dt} \int_{CV} \rho \vec{V} dV + \int_{CS} \rho \vec{V} (\vec{V} \cdot \vec{n}) dA$

1-D: $\Sigma F = \frac{\dot{m}}{gc} (V_o - V_i)$

$\frac{dP}{\rho} + \frac{V^2}{2gc} \frac{dV}{V} + \frac{g}{gc} dz + \frac{dV^2}{2gc}$

$T ds_i = \int \frac{V^2}{2gc} \frac{dV}{V}$ $\delta = \frac{4\mu V}{D \rho V_c}$ $Re = \frac{\rho V D}{\mu}$

Oblique Shock



$\tan(\delta) = 2 \cot(\theta) \left(\frac{M_1^2 \sin^2 \theta - 1}{M_1^2 (\gamma + \cos(2\theta)) + 2} \right)$

$M_{1n} = M_1 \sin(\theta)$

$M_{2n} = \frac{M_{1n}}{\sin(\theta - \delta)}$

$P_3 > P_{amb} > P_2$ $P_4 = P_6 = P_{amb}$

$\gamma = \sqrt{\frac{\gamma+1}{\gamma-1} (M^2 - 1)}$

$\gamma = \sqrt{\frac{\gamma+1}{\gamma-1} (M^2 - 1)} - \tan^{-1} \sqrt{M^2 - 1}$

Prandtl-Meyer Flow

$V_t = V \cos \mu$ $d\mu = \sqrt{M^2 - 1} \frac{dV}{V}$

$\mu = \sqrt{\frac{\gamma+1}{\gamma-1}} \tan^{-1} \left(\sqrt{\frac{\gamma-1}{\gamma+1} (M^2 - 1)} \right) - \tan^{-1} \sqrt{M^2 - 1}$

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2nd: $\Delta S \int \frac{\delta q}{T} \Rightarrow ds = \frac{\delta q}{T}$

$\delta q = \delta w + du \Rightarrow T ds = du + P dV$

Entropy Changes $ds_{1-2} = C_p \ln\left(\frac{T_2}{T_1}\right) + C_v \ln\left(\frac{P_2}{P_1}\right)$

$ds_{1-2} = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right)$, $ds_{1-2} = C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{P_2}{P_1}\right)$

Sonic Speed $a^2 = \gamma c_p \left(\frac{RT}{M}\right)$, $M = \frac{V}{a}$ $\sin \mu = \frac{1}{M}$

Special Relations for Perfect Gas

$a^2 = \gamma c_p R T$ $h_t = h \left(1 + \frac{\gamma-1}{2} M^2\right)$ $T_t = T \left(1 + \frac{\gamma-1}{2} M^2\right)$

$P_t = P \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}}$ $\frac{dP_t}{P_t} + \frac{dS_c}{R} \left(1 - \frac{\gamma}{\gamma-1}\right) + \frac{dS_i}{R} + \frac{dV^2}{2gc} = 0$

$\frac{P_t}{P} = e^{-\Delta s_i/R}$ for $Q = W = 0$

S. 1-D flow of Perfect Gas when $Q = W = 0$

$\frac{P_{t1}}{P_{t2}} = \frac{A_2^*}{A_1^*} = e^{-\Delta s_i/R} \Rightarrow P_{t1} A_1^* = P_{t2} A_2^*$

Nozzle Performance $\eta_n = \frac{\Delta h_{actual}}{\Delta h_{ideal}} = \frac{h_1 - h_{t2}}{h_1 - h_{t2s}}$

Diffuser Performance $\eta_d = \frac{P_{t2}}{P_{t1}}$

Total Pressure Recovery or diffuser efficiency (same definition) $\eta_d = \frac{\Delta P}{\Delta P_s} = \frac{P_{t2} - P_2}{P_{t2} - P_1}$

General fluid No losses $\frac{dP}{\rho} = \frac{P V^2}{\rho} \left(\frac{1}{1-M^2}\right) \frac{dA}{A}$ $\frac{dP}{P} = \left(\frac{M^2}{1-M^2}\right) \frac{dA}{A}$

$\frac{dV}{V} = -\left(\frac{1}{1-M^2}\right) \frac{dA}{A}$ $\frac{dP}{P} = -M^2 \frac{dV}{V}$

Perfect Gas w/ losses $\frac{A_2}{A_1} = \frac{M_1}{M_2} \left(\frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_1^2}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$

$\frac{T_2}{T_1} = \frac{1 + \frac{\gamma-1}{2} M_1^2}{1 + \frac{\gamma-1}{2} M_2^2}$

Isentropic Working Eq $\frac{T}{T_t} = \frac{1}{1 + \frac{\gamma-1}{2} M^2}$ $\frac{P}{P_t} = \left(\frac{T}{T_t}\right)^{\frac{\gamma}{\gamma-1}}$

$\frac{A}{A^*} = \frac{1}{M} \left(\frac{1 + \frac{\gamma-1}{2} M^2}{\frac{\gamma+1}{2}}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$

Nozzle Operation $\frac{\dot{m}}{A} = \frac{P_t}{\sqrt{T_t}} \sqrt{\frac{\gamma}{R}} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2}}$

$h_{t1} = h_{t2}$, $T_{t1} = T_{t2}$ $P_2 = P_{rec}$

Shock General fluid $h_{t1} = h_{t2} \Rightarrow h_1 + \frac{V_1^2}{2gc} = h_2 + \frac{V_2^2}{2gc}$

$P_1 V_1 = P_2 V_2$ $\Sigma F_x \Rightarrow \Delta P = \frac{\rho \Delta V^2}{2} = \frac{\rho V_1^2 - \rho V_2^2}{2} \Rightarrow P_1 + \frac{\rho V_1^2}{2} = P_2 + \frac{\rho V_2^2}{2}$

General Working Eq Perfect Gas

$\frac{P_1 M_1}{\sqrt{T_1}} = \frac{P_2 M_2}{\sqrt{T_2}}$ $T_1 \left(1 + \frac{\gamma-1}{2} M_1^2\right) = T_2 \left(1 + \frac{\gamma-1}{2} M_2^2\right)$

$P_1 \left(1 + \gamma M_1^2\right) = P_2 \left(1 + \gamma M_2^2\right)$ $\frac{V_1 - V_2}{a_1} = \left(\frac{2}{\gamma+1}\right) \left(\frac{M_1^2 - 1}{M_1}\right)$

$\frac{V_1 - V_2}{a_1} = \left(\frac{2}{\gamma+1}\right) \left(\frac{M_1^2 - 1}{M_1}\right)$

Normal Velocity Squar Position: Moving Normal Shock

$P_1 = P_2 = P'$ $T_1 = T_2 = T'$ $V_1 = V_2 = V'$

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$\delta = \frac{C_p}{C_v}$, $C_p - C_v = \frac{R}{M}$

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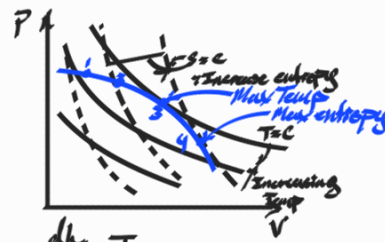
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Rayleigh Flow Steady 1-D, $h_c = 0$, $ds = 0$, $dM = 0$
 $ds_i \ll ds_c \rightarrow ds_i \approx 0 \rightarrow G = \text{const}$
 $T_c \neq \text{constant}$, $p + G^2 V = \text{const}$

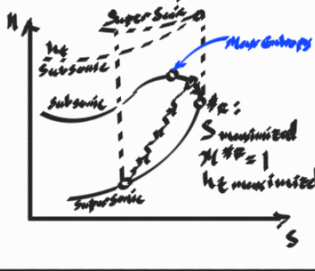
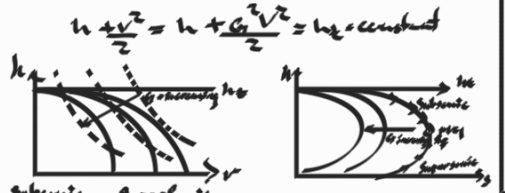
Subsonic	Supersonic	Properties
Upper	Lower	
$T \uparrow$	$T \downarrow$	h
$p \uparrow$	$p \downarrow$	P
$V \downarrow$	$V \uparrow$	V
$\rho \downarrow$	$\rho \uparrow$	ρ
$\mu \downarrow$	$\mu \uparrow$	μ



Variable Area Flow		
$dA > 0$	$M > 1$	$M < 1$
$P \downarrow$	$P \downarrow$	$P \uparrow$
$T \downarrow$	$T \downarrow$	$T \uparrow$
$V \uparrow$	$V \uparrow$	$V \downarrow$

$dA > 0$	$M < 1$	$M > 1$
$P \uparrow$	$P \uparrow$	$P \downarrow$
$T \uparrow$	$T \uparrow$	$T \downarrow$
$V \downarrow$	$V \downarrow$	$V \uparrow$

Euro flow
 Constant Area, force of friction
 Constant $m = \rho A V \rightarrow \rho V = \text{const}$
 $\rightarrow h_c = \text{constant}$



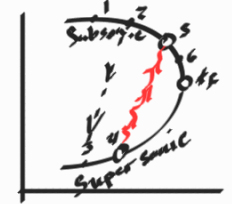
Working Equations
 From Mass eq: $\frac{P_2}{P_1} = \frac{1 + \gamma M_1^2}{1 + \gamma M_2^2}$
 From Continuity: $\frac{T_2}{T_1} = \left(\frac{1 + \gamma M_1^2}{1 + \gamma M_2^2} \right)^{\frac{\gamma}{\gamma-1}}$

Nozzle & Diffuser
 where:
 $M > 1$ is nozzle
 $M < 1$ is diffuser
 converts $h + K$
 converts K to h
 C-D Nozzles
 $M < 1$ and $M > 1$
 $M > 1$ and $M < 1$
 acts as diffuser.

Subsonic	Supersonic	Properties
Upper	Lower	
$T \downarrow$	$T \uparrow$	h
$p \downarrow$	$p \downarrow$	P
$V \downarrow$	$V \uparrow$	V
$\rho \downarrow$	$\rho \uparrow$	ρ
$\mu \downarrow$	$\mu \uparrow$	μ

Working Equations
 $h_{c1} = h_{c2}$
 $T_{c1} = T_{c2}$
 $T_c = \text{constant}$

Stagnation
 $\frac{T_{c2}}{T_{c1}} = \left(\frac{1 + \gamma M_1^2}{1 + \gamma M_2^2} \right)^{\frac{\gamma}{\gamma-1}}$
 $\frac{P_{c2}}{P_{c1}} = \left(\frac{1 + \gamma M_1^2}{1 + \gamma M_2^2} \right)^{\frac{\gamma}{\gamma-1}}$
 From NR eq: $q = C_p \Delta T_c \neq C_p \Delta T$

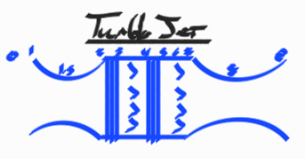
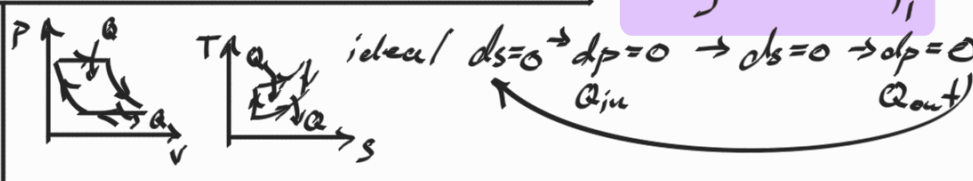


Reference States
 $\frac{P}{P_1} = \frac{1 + \gamma}{1 + \gamma M_1^2}$, $\frac{T}{T_1} = \frac{M_1^2 (1 + \gamma)}{(1 + \gamma M_1^2)^2}$
 $\frac{T_c}{T_1} = \frac{2(1 + \gamma) M_1^2}{(1 + \gamma M_1^2)^2}$
 $\frac{P_c}{P_1} = \left(\frac{1 + \gamma}{1 + \gamma M_1^2} \right) \left(\frac{2 M_1^2}{1 + \gamma M_1^2} \right)^{\frac{\gamma}{\gamma-1}}$
 $\frac{S_{max}}{R} = \frac{\gamma-1}{\gamma} \ln \left(\frac{1}{M_1^2} \right) + \frac{\gamma+1}{2(\gamma-1)} \ln \left(\frac{P_1}{P} \right)$

Propulsion efficiencies
 $\eta_{\text{Brayton}} = 1 - \frac{T_0}{T_1}$

Good for $M_0 \sim 3.5$
 Bad in both
 $M > 5$ and Lower Mach
 $\eta_{\text{ideal}} = 1 - \frac{T_0}{T_{c0}}$

$T_0 = \frac{T_{\text{out}}}{T_{\text{in}}}$
 $\eta = \frac{P_{\text{out}}}{P_{\text{in}}}$
 $T_{c1} = \frac{T_{c0}}{1 + \frac{\gamma-1}{2} M_0^2}$
 $\eta_r = \frac{P_{c1}}{P_0} = \eta_r$
 $T_{c1} = \frac{T_{c0}}{T_0}$



Ramjet flow
ideal

Engine Performance Parameters
 NRQ Balance across the engine
 $\Rightarrow (\dot{m}_a + \dot{m}_f) h_{c2} = \dot{m}_a h_{c1} + \dot{m}_f h_f$
 Capture Area
 $\Rightarrow \dot{m}_a = \rho_0 V_0 A_0 = G_0 A_0$

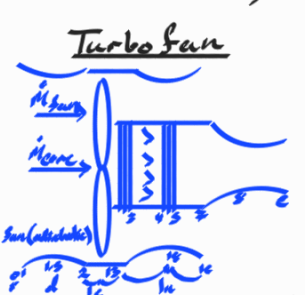
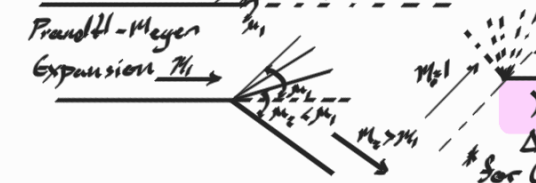
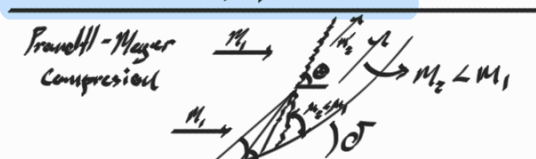
$\eta_{pr} = \frac{T V_0}{\dot{m}_a \left(\frac{V_c^2}{2} - \frac{V_0^2}{2} \right) + \dot{m}_f \frac{V_c^2}{2}} \approx \frac{2 V_0}{V_0 + V_c} \mid P_c = P_0$
 $\eta_{th} = 1 - \frac{Q_{\text{rejected}}}{Q_{\text{in}}}$
 $\eta_{ov} = \frac{T V_0}{\dot{m}_f h_f} = \eta_{pr} \eta_{th}$
 $I_{sp} = \frac{T}{\dot{m}_f g} [s]$
 $SFC = \frac{3600}{I_{sp}}$

Good for $M_0 < 1, \sim 1$
 More compression for lower M_0
 $\eta_{\text{ideal}} = 1 - \frac{T_{c0}}{T_{c3}}$

$\dot{m} = G A = \frac{\gamma}{(\frac{\gamma+1}{2})^{\frac{\gamma+1}{2(\gamma-1)}}} \left(\frac{P_c A}{\sqrt{\gamma P_c}} \right) f(M)$

$f(M) = \frac{A^*}{A} = \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \frac{M}{(1 + \frac{\gamma-1}{2} M^2)^{\frac{\gamma+1}{4(\gamma-1)}}}$

From Conservation of $\dot{m}$: (example)
 $\frac{P_{c2}}{\sqrt{T_{c2}}} f(M_2) = \frac{P_{c1}}{\sqrt{T_{c1}}} f(M_1)$



Turbofan
 Lower M_0 (commercial)
 $\beta = \frac{\dot{m}_{fan}}{\dot{m}_{core}}$
 $B = \frac{\dot{m}_{fan}}{\dot{m}_{fan} + \dot{m}_{core}}$
 $\eta_{\text{ideal}} = 1 - \frac{T_0}{T_{c3}}$